https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Table of Contents

Introduction 1

About This Book 1
Conventions Used in This Book 1
Foolish Assumptions 2
How This Book Is Organized 2
Part I: ICND1 – Exam 100-101 2
Part II: ICND2 – Exam 200-101 3
Beyond the Book 3
What you'll find online 3
How to register 3
Extra content 4
The Value of CCNA Certification 4
CCNA Exam Details 5
What to Expect on the CCNA Exams 6
Types of questions 6
Exam objectives 7
On Your Test Day 8
Arriving at the test site 8
Getting lots of rest 8
Answering questions 8

Part I: ICND 1 – Exam 100-101 9

Chapter 1: Networking Devices, Technologies, and Models 11

The Questions You'll Work On 11
What to Watch Out For 11
Network Services 12
Network Devices and Communication 13
OSI Model and Network Standards 15
Network Cabling 18

Chapter 2: Introduction to TCP/IP 21

The Questions You'll Work On 21
What to Watch Out For 21
IP Addressing 22
TCP/IP Protocols 25
IPv6 28
TCP/IP Ports 29
Troubleshooting 30

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Chapter 3: Subnetting and VLSM 33

The Problems You'll Work On 33
What to Watch Out For 33
Subnetting 34
VLSM 41

Chapter 4: Introduction to Cisco IOS 45

The Problems You'll Work On 45
What to Watch Out For 45
Device Ports 46
Memory Types 48
IOS Basics 50
Boot Process 52
IOS Version Information 54

Chapter 5: Basic Router Configuration 57

The Problems You'll Work On 57
What to Watch Out For 57
Configuring Interfaces 58
Basic Configuration 60
Basic Troubleshooting 63

Chapter 6: Managing Cisco Devices 69

The Problems You'll Work On 69
What to Watch Out For 69
Backup and Remote 70
Cisco Discovery Protocol (CDP) 73
Telnet 76

Chapter 7: Advanced Router Topics 79

The Problems You'll Work On 79
What to Watch Out For 79
Hostname Resolution 80
Configuration of DHCP Service 82
Implementing NAT 83
Password Recovery 86
Access Control Lists (ACLs) 88

Chapter 8: Static Routing 91

The Problems You'll Work On 91
What to Watch Out For 91
Introduction to Routing and Communication 92
Configuring Static Routes 94
Gateway of Last Resort 96
Troubleshooting Connectivity 97
Router on a Stick 100
Administrative Distances 101

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Chapter 9: Dynamic Routing Protocols 103

The Problems You'll Work On 103
What to Watch Out For 103
Introduction to Dynamic Routing Protocols 104
Configuring RIP and RIPv2 107
Basics of Configuring OSPF 110

Chapter 10: Introduction to Switching 113

The Problems You'll Work On 113
What to Watch Out For 113
Introduction to Cisco Switches 114
Understanding Switch Functionality 115
Understanding Data Flow 120
Basic Switch Configuration 122

Chapter 11: Basic Switch Configuration 125

The Problems You'll Work On 125
What to Watch Out For 125
Configuring Ports 126
Troubleshooting Switch Configuration 128
Port Security 131
Configuring VLANs 134

Chapter 12: Troubleshooting Network Communication 137

The Problems You'll Work On 137
What to Watch Out For 137
Troubleshooting Commands 138
Troubleshooting Connectivity Issues 140
Using Show Commands to Troubleshoot 145
Using Debug Commands 151

Chapter 13: Cisco Device Security Best Practices 153

The Problems You'll Work On 153
What to Watch Out For 153
Security Basics 154
Security Configuration 155
Switch Security 160
Configuring Passwords 163
Configuring Banners 165
Remote Management 166

Chapter 14: Introduction to Wireless Networking 169

The Problems You'll Work On 169
What to Watch Out For 169
Wireless Basics 170
Wireless Security 175
Wireless Standards 177

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Chapter 15: Introduction to WANs 179

The Problems You'll Work On 179
What to Watch Out For 179
Introduction to WAN Terminology 180
Methods of Connecting to WANs 182
Configuring Serial Links 183

Part II: ICND 2 – Exam 200-101 191

Chapter 16: ICND1 Review 193

The Problems You'll Work On 193
What to Watch Out For 193
Networking Basics 194
IP Addressing, Subnetting and VLSM 197
Configuring Routers and IP Routing 202
Configuring Switches 213
Troubleshooting and Security 218
Wireless and WANs 223

Chapter 17: Switching Technologies 227

The Problems You'll Work On 227
What to Watch Out For 227
Understanding Spanning Tree Protocol (STP) 228
Configuring VLANs 232
Configuring VLAN Trunking Protocol (VTP) 236
Implementing Etherchannel and Port Security 239

Chapter 18: Cisco IOS Fundamentals 241

The Problems You'll Work On 241
What to Watch Out For 241
Understanding the Cisco IOS Boot Process 242
Working with IOS Images 245
Managing Cisco IOS Files 247
Working with Licenses 250

Chapter 19: Router Essentials and Route Summarization 253

The Problems You'll Work On 253
What to Watch Out For 253
Understanding Static and Dynamic Routing 254
Recognizing Routing Terminology 259
Working with Route Summarization 260
Knowing about Inter-VLAN Routing 261

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Chapter 20: RIP and OSPF 265

The Problems You'll Work On 265
What to Watch Out For 265
Understanding RIP 266
Configuring RIP 267
Understanding OSPF 271
Configuring OSPF 273

Chapter 21: Routing with EIGRP 277

The Problems You'll Work On 277
What to Watch Out For 277
Understanding EIGRP 278
Configuring EIGRP 281
Troubleshooting EIGRP 284

Chapter 22: Configuring IP Services 289

The Problems You'll Work On 289
What to Watch Out For 289
Reviewing Various High Availability (HA) Options 290
Managing and Using Syslog Servers 295
Understanding SNMP v2 and v3 297
Accessing Statistics Using NetFlow 299

Chapter 23: Frame Relay 301

The Problems You'll Work On 301
What to Watch Out For 301
Understanding Frame Relay Basics 302
Configuring Frame Relay 303
Implementing Frame Relay 304
Troubleshooting Frame Relay 306

Chapter 24: WAN Technologies 311

The Problems You'll Work On 311
What to Watch Out For 311
Configuring Serial Ports 312
Understanding Frame Relay 315
Troubleshooting WAN Connections 317

Part III: The Answers 323

Answers 325

Index 449